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ORIGINAL ARTICLES January 13, 2017 • 1 Comment

Introduction to Econometrics

Today, we live in a world where information is at one's disposal with the single click of a button. The result of this is a growing demand for methods in understanding, analyzing, and modelling data. In

economics, we refer to the development and usage of statistical techniques as econometrics.

Econometrics begins with an economic question: a relationship in which we are interested in studying. This could be a theory we wish to prove, or a policy's effects we are trying to understand. Once we have posed a question, we can hypothesize a model that we believe would capture the relationship (Wooldridge 2). For example, consider the economic question, "What affects a person's wage rate?". Suppose that we believe education is a factor, and relationship is captured with the equation:

$$\text{Wage} = f(\text{Education})$$

Perhaps to most of us, this follows quite naturally, despite not having done any form of analysis. However, an econometrician will often tell you this is not the case and a deeper study of this question is required.

From an econometric perspective, the above equation fails to answer two crucial questions:

- What is the magnitude in which education affects wage?
- Is education the only factor in determining wage?

Luckily, we can quickly transform our equation into an econometric regression model.

$$\text{Wage}_i = \beta_0 + \beta_1 \text{Education}_i + \epsilon_i$$

By collecting enough data (sets of information), we are able to conduct an empirical analysis to determine the estimates for the parameters β_0 and β_1 . The result is a relationship explained through a numerical equation, backed by a set of observations. With statistical tests, we can also assess the strength of our model. If it is weak, we

know that some important factors may be missing, such as age. We can continue this process until an optimal model is achieved.

While econometric methods do carry enormous predictive and analytical power, they cannot be used indiscriminately. In economics, we are highly interested in causality through *ceteris paribus*: Keeping all other things equal (Wooldridge 12). However, it is often difficult to create controlled experiments to achieve this concept. Thus, when creating econometric models, it is imperative that we preserve *ceteris paribus* when selecting our sample of data. For example, assume the government made a policy change so that families with under \$50,000 household income would receive a subsidy for their child's education. We could measure the effects of this subsidy by observing the same group of families through the policy change. However, if different samples of families were drawn before and after the subsidy, we may inadvertently include the variation across households in addition to the effects of the policy change.

Another area to consider is the interpretation of statistical correlation. Sometimes, econometric models may report substantial correlation between two variables. In particular, time-series data (variables that change over time) often exhibit correlation if not corrected for when estimating. Indeed, statistical correlation does not imply causation. For example, consider number of pregnancies and number of doctors in a city. There may be data that shows strong correlation between the two variables, but we know that this is likely a coincidence and not a causal effect.

Econometrics uses a set of statistical tools in economic settings to derive conclusions and empirical results. With the prolific demand for studying data, it is important for us as economists to understand this field of study. In a society surrounded by data, econometrics is our key to better understand and observe the world.

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REPLY 

[...] In the previous article, we introduced the motivation behind econometrics and the role it plays in the field of economics. We also briefly discussed the concept of an econometric model, which was essentially an equation that captures the relationship between variables. Today we will leap further into the econometric discussion by examining the most fundamental model: Simple Linear Regression (SLR). [...]

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